

nano input.txt

hello pict hello
pict hadoop pict

nano WordCount.java

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable>{
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();
        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {
            StringTokenizer itr = new StringTokenizer(value.toString());
            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer
        extends Reducer<Text, IntWritable, Text, IntWritable>{
        public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {
            int
sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
        }
    }
}
```

```

        context.write(key, new IntWritable(sum));
    }
}
public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Word Count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

omkar@LAPTOP-S6G2S414:/mnt/c/Users/Omkar.LAPTOP-S6G2S414\$ start-
dfs.sh

Starting namenodes on [localhost]

Starting datanodes

Starting secondary namenodes [LAPTOP-S6G2S414]

omkar@LAPTOP-S6G2S414:/mnt/c/Users/Omkar.LAPTOP-S6G2S414\$ start-
yarn.sh

Starting resourcemanager

Starting nodemanagers

omkar@LAPTOP-S6G2S414:~/wordcount\$ export
HADOOP_CLASSPATH=\$(hadoop classpath)

omkar@LAPTOP-S6G2S414:~/wordcount\$ javac -classpath
"\$HADOOP_CLASSPATH" -d . WordCount.java

omkar@LAPTOP-S6G2S414:~/wordcount\$ jar -cvf WordCount.jar *.class
added manifest
adding: WordCount\$IntSumReducer.class(in = 1657) (out= 700)(deflated 57%)

adding: WordCount\$TokenizerMapper.class(in = 1752) (out= 764)(deflated 56%)

adding: WordCount.class(in = 1472) (out= 806)(deflated 45%)

```
omkar@LAPTOP-S6G2S414:~/wordcount$ hdfs dfs -mkdir -p /input
hdfs dfs -put input.txt /input
```

```
omkar@LAPTOP-S6G2S414:~/wordcount$ hdfs dfs -ls /input
```

Found 1 items

```
-rw-r--r--  1 omkar supergroup      34 2026-04-21 20:15 /input/input.txt
```

```
omkar@LAPTOP-S6G2S414:~/wordcount$ hadoop jar WordCount.jar
```

```
WordCount /input /output
```

```
2026-04-21 20:20:26,435 INFO client.DefaultNoHARMFailoverProxyProvider:
Connecting to ResourceManager at /0.0.0.0:8032
```

```
2026-04-21 20:20:26,900 WARN mapreduce.JobResourceUploader: Hadoop
command-line option parsing not performed. Implement the Tool interface and
execute your application with ToolRunner to remedy this.
```

```
2026-04-21 20:20:26,943 INFO mapreduce.JobResourceUploader: Disabling
Erasure Coding for path: /tmp/hadoop-
yarn/staging/omkar/.staging/job_1776802751232_0002
```

```
2026-04-21 20:20:27,910 INFO input.FileInputFormat: Total input files to
process : 1
```

```
2026-04-21 20:20:28,019 INFO mapreduce.JobSubmitter: number of splits:1
```

```
2026-04-21 20:20:28,269 INFO mapreduce.JobSubmitter: Submitting tokens for
job: job_1776802751232_0002
```

```
2026-04-21 20:20:28,269 INFO mapreduce.JobSubmitter: Executing with
tokens: []
```

```
2026-04-21 20:20:28,444 INFO conf.Configuration: resource-types.xml not
found
```

```
2026-04-21 20:20:28,444 INFO resource.ResourceUtils: Unable to find
'resource-types.xml'.
```

```
2026-04-21 20:20:28,744 INFO impl.YarnClientImpl: Submitted application
application_1776802751232_0002
```

```
2026-04-21 20:20:28,826 INFO mapreduce.Job: The url to track the job:
http://LAPTOP-S6G2S414.:8088/proxy/application_1776802751232_0002/
```

```
2026-04-21 20:20:28,827 INFO mapreduce.Job: Running job:
job_1776802751232_0002
```

```
2026-04-21 20:20:37,995 INFO mapreduce.Job: Job job_1776802751232_0002
running in uber mode : false
```

2026-04-21 20:20:37,996 INFO mapreduce.Job: map 0% reduce 0%
2026-04-21 20:20:43,076 INFO mapreduce.Job: map 100% reduce 0%
2026-04-21 20:20:49,141 INFO mapreduce.Job: map 100% reduce 100%
2026-04-21 20:20:50,163 INFO mapreduce.Job: Job job_1776802751232_0002
completed successfully

2026-04-21 20:20:50,287 INFO mapreduce.Job: Counters: 54

File System Counters

FILE: Number of bytes read=76
FILE: Number of bytes written=551839
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=136
HDFS: Number of bytes written=24
HDFS: Number of read operations=8
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

Job Counters

Launched map tasks=1
Launched reduce tasks=1
Data-local map tasks=1
Total time spent by all maps in occupied slots (ms)=3000
Total time spent by all reduces in occupied slots (ms)=3414
Total time spent by all map tasks (ms)=3000
Total time spent by all reduce tasks (ms)=3414
Total vcore-milliseconds taken by all map tasks=3000
Total vcore-milliseconds taken by all reduce tasks=3414
Total megabyte-milliseconds taken by all map tasks=3072000
Total megabyte-milliseconds taken by all reduce tasks=3495936

Map-Reduce Framework

Map input records=2
Map output records=6
Map output bytes=58
Map output materialized bytes=76
Input split bytes=102
Combine input records=0
Combine output records=0
Reduce input groups=3
Reduce shuffle bytes=76

Reduce input records=6
Reduce output records=3
Spilled Records=12
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=67
CPU time spent (ms)=1360
Physical memory (bytes) snapshot=510603264
Virtual memory (bytes) snapshot=5507362816
Total committed heap usage (bytes)=392167424
Peak Map Physical memory (bytes)=283725824
Peak Map Virtual memory (bytes)=2750844928
Peak Reduce Physical memory (bytes)=226877440
Peak Reduce Virtual memory (bytes)=2756517888
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=34
File Output Format Counters
Bytes Written=24

```
omkar@LAPTOP-S6G2S414:~/wordcount$ hdfs dfs -cat /output/part-r-00000
hadoop 1
hello 2
pict 3
```